RSGT CHITTAGONG
Part of RSGT Bangladesh

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Case Study 2: Predictive Workforce Planning (CV Delivery Forecasting)

Background

RSGT Bangladesh handles two primary types of container deliveries:

1. **On-Chassis Delivery (OCD):** The full container is loaded onto a customer's trailer.
2. **Covered Van (CV) Delivery:** Cargo is "unstripped/Unloaded" from the container and loaded into covered vans. This process is labor and equipment intensive.

Currently, CV delivery volumes fluctuate significantly, making it difficult for the Terminal Manager to allocate the correct number of laborers and forklifts. Under-staffing leads to congestion, while over-staffing leads to high operational costs.

The Challenge

You are tasked with predicting the daily number of **CV Delivery for the first 10 days of February'26** based on 5 months of historical data.

Key Operational Factors:

- **The 4-Day Rule:** Port regulations allow 4 "Free Days" for storage. Historically, delivery probability peaks on the **3rd and 4th day** after the container enters the yard.
- **Weekly Seasonality:** Delivery volumes vary by the day of the week (e.g., weekends vs. weekdays).
- **Yard Density Factor:** There is a known correlation between the total inventory in the yard and the number of CV deliveries requested.

February Import Scenario:

The number of CV imports on first 10 days of February was:

157	59	0	0	0	184	319	221	87	115

On 1st September the yard had 300 CV containers.

Technical Tasks

Task : Prediction, Methodology & Justification

- Predict the CV delivery for first 10 days of February.
- Provide a brief technical note explaining your chosen model for prediction (e.g., Moving Average, Linear Regression, or Time-Series ML models).
- Justify why you have selected this method.

Deliverables

1. **The Prediction Report (PDF):**
   - The forecasted numbers for the next 10 days.
   - A summary of your methodology and assumptions.
2. **Working Files:** Your Excel model, or Script used to generate the results.

Note:

If you feel any information is missing, make a logical assumption and mention it in your deliverables.

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